1)

a) According to the following equations:

$Z_{k1} = s_1Z_{12} + s_2Z_{23} + s_3Z_{13}$; $Z_{k2} = s_1Z_{13} + s_2Z_{12} + s_3Z_{23}$

$Z_{k3} = s_1Z_{23} + s_2Z_{13} + s_3Z_{12}$; $Z_S = s_1Z_{11} + s_2Z_{22} + s_3Z_{33}$

Plug in the values from the matrix above:

Z_{k1} = .4*(3.812 + *j*27.332) + .4*(3.812 + *j*27.332) + .2*(3.812 + *j*23.968) = **3.812 + *j*26.6592**

Z_{k2} = .4*(3.812 + *j*23.968) + .4*(3.812 + *j*27.332) + .2*(3.812 + *j*27.332) = **3.812 + *j*26.9864**

Z_{k3} = .4*(3.812 + *j*27.332) + .4*(3.812 + *j*27.968) + .2*(3.812 + *j*23.332) = **3.812 + *j*26.9864**

Z_S = .4*(14.932 + *j*58.376) + .4*(14.932 + *j*58.376) + .2*(14.932 + *j*58.376) = **14.932 + *j*58.376**

b) Using the following equations:

$Z_{k1} = s_1Z_{12} + s_2Z_{13}$; $Z_{k2} = s_1Z_{13} + s_2Z_{12}$

$Z_{k3} = s_1Z_{23} + s_2Z_{23} = Z_{23}$; $Z_S = Z_{11} = Z_{22} = Z_{33}$

Plug in the values from the matrix above:

Z_{k1} = .5*(3.812 + *j*27.332) + .5*(3.812 + *j*23.968) = **3.812 + *j*26.65**

Z_{k2} = .5*(3.812 + *j*23.968) + .5*(3.812 + *j*27.332) = **3.812 + *j*26.65**

Z_{k3} = .5*(3.812 + *j*27.332) + .5*(3.812 + *j*27.332) = **3.812 + *j*27.332**

Z_{k3} = .4*(14.932 + *j*58.376) + .4*(14.932 + *j*58.376) + .2*(14.932 + *j*58.376) = **14.932 + *j*58.376**

2)

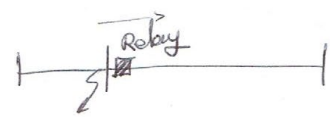
To determine the direction of the fault, the following equations can be used

$$S_{1R} = \Delta V_R - Z_0 \Delta i_R$$
$$S_{2R} = \Delta V_R + Z_0 \Delta i_R$$

where

$$\Delta V_R = \Delta V_R^+ + \Delta V_R^-$$
$$\Delta i_R = \Delta i_R^+ + \Delta i_R^-$$

* In the case of reverse faults (i.e. fault behind the Relay)



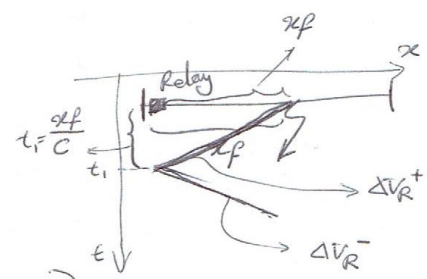
$$\Delta V_R^- = \Delta i_R^- = 0$$
$$\Delta V_R^+ = 1\,pu$$
$$\Delta i_R^+ = \frac{\Delta V_R^+}{Z_0} = \frac{1}{Z_0}$$

$$\Delta V_R = 1 + 0 = 1$$
$$\Delta i_R = \frac{1}{Z_0} + 0 = \frac{1}{Z_0}$$

$$\boxed{\begin{aligned} S_{1R} &= 1 - Z_0 \times \frac{1}{Z_0} = 0 \\ S_{2R} &= 1 + Z_0 \times \frac{1}{Z_0} = 2\,pu \end{aligned}}$$

1

* In the case of forward faults



$$\Delta V_R^+ = \Delta V_f \left(t - \frac{x_f}{c}\right)$$
$$\Delta i_R^+ = -\frac{\Delta V_f \left(t - \frac{x_f}{c}\right)}{Z_0}$$

$$\Delta V_R^- = K V_R^+$$
$$\Delta i_R^- = K \Delta i_R^+$$
$$K = \frac{Z_{0R} - Z_0}{Z_{0R} + Z_0} = \frac{\frac{Z_0}{2} - Z_0}{\frac{Z_0}{2} + Z_0} = -\frac{1}{3}$$

$$\Delta V_R^- = -\frac{\Delta V^+}{3}$$
$$\Delta i_R^- = -\frac{\Delta i_R^+}{3}$$

$$\Delta V_R = \Delta V_R^{+} + \Delta V_R^{-}$$

$$= \Delta V_f\left(t - \frac{x_f}{c}\right) - \frac{Z_0}{3}\Delta V_f\left(t - \frac{x_f}{c}\right) = \frac{2}{3}\Delta V_f\left(t - \frac{x_f}{c}\right)$$

$$\Delta i_R = \Delta i_R^{+} + \Delta i_R^{-}$$

$$= -\frac{\Delta V_f\left(t - \frac{x_f}{c}\right)}{Z_0} - \frac{\Delta V_f\left(t - \frac{x_f}{c}\right)}{3Z_0} = -\frac{4\,\Delta V_f\left(t - \frac{x_f}{c}\right)}{3Z_0}$$

$$\left.\begin{aligned} S_{1R} &= \frac{2}{3}\Delta V_f\left(t - \frac{x_f}{c}\right) + \frac{4}{3}\Delta V_f\left(t - \frac{x_f}{c}\right) = 2\Delta V_f\left(t - \frac{x_f}{c}\right) \\ S_{2R} &= \frac{2}{3}\Delta V_f\left(t - \frac{x_f}{c}\right) - \frac{4}{3}\Delta V_f\left(t - \frac{x_f}{c}\right) = -\frac{2}{3}\Delta V_f\left(t - \frac{x_f}{c}\right) \end{aligned}\right\}$$

$$\Delta V_f = 1\,\text{pu}$$

$$\boxed{\begin{aligned} S_{1R} &= 2\,\text{pu} \\ S_{2R} &= -\frac{2}{3}\,\text{pu} \end{aligned}} \quad (2)$$

According to (1) and (2), in the case of forward faults S_{1R} and S_{2R} become non-zero simultaneously. However, in the case of reverse faults S_{1R} and S_{2R} do not become non-zero simultaneously.